

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Pipeline Construction Plan Inclusive of WPS/PQR
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Pipeline Construction Plan Inclusive of WPS/PQR

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Revision History

Rev No.	Date	Description of Revision
R01	09.08.22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-PLN-003

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design

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Definition of Terms

Terms	Definitions
Audit	Systematic, independent and documented process for obtaining audit evidence and evaluating it objectively to determine the extent to which audit criteria are fulfilled (ISO 9000 "Quality Management Systems – Fundamentals and Vocabulary"). To be considered: Regular Audit: Audit of an activity or a vendor carried out with regular planned intervals Spot Audit: Planned assessment of a specific activity or a selective audit requested additionally to the agreed / approved schedule. All audits are planned and referenced in the agreed / approved Project Audit Schedule
BFPCL	Brass Fertilizer & Petrochemical Limited.
Calibration	Comparison of a measurement and test equipment with a reference standard or with another one provided with better tolerance in order detect or to measure possible inaccuracies and to document or eliminate them.
Company	Brass Fertilizer & Petrochemical Ltd
Concession Request	Concession Request is a written request to COMPANY for accepting a non-conforming product, and/or any other else occurred deviations of product / services rendered by CONTRACTOR against COMPANY contractual requirements, in the event CONTRACTOR considers there are sound reasons for COMPANY to accept related deviations.
Consultant	Project Management Consultant
Contractor	Epic Atlantic Ltd
Criticality Element	Criticality Element gives a project specific classification of the equipment and material based on geographical location and the cost risk involved.
Criticality Rating	Based on the Product Category & Criticality Element Analysis, the Criticality Rating gives the minimum Inspection Level, (and Inspection Class as per SAP software) to be adopted.
Deviation Request	During detailed design and fabrication activities, requests to deviate from specified requirements, approved design drawings, specifications, applicable norms, codes, and standards as well as incorporation into the works of nonconforming products, in the event that CONTRACTOR considers there to be sound reasons for COMPANY to accept, shall be made through a written deviation request to grant a concession to the Contract requirements using the agreed deviation request form.
Employer	Party (Usually BFPCL) that initiates the project and ultimately pays for its

	design and construction. The Employer will generally specify the technical requirements. The Employer may also include an agent or consultant authorized to act for and on behalf of, the Employer.
Engineer	The party that carries out the FEED Engineering of the project.
Facility	Facility, associated utilities, machinery, apparatus, article and supporting infrastructure to be constructed and warranted in accordance with the contract
Inspection and Test Plan	CONTRACTOR vendor / subcontractor plans prepared in accordance with the corresponding ITS and applicable contractual, engineering and vendors requirements. The ITP shall provide references to applicable procedures, acceptance criteria, documentation, quality records and inspection attendance to be performed, in chronological order, from initial inspection, throughout final inspections and mechanical completion. ITP(s) and relevant inspection report forms shall be enclosed in the document.
Manufacturer/Supplier	Party, which manufactures or supplies equipment and services to perform the duties specified by the Contractor.
Measuring Range	Range defined by LRL and URL within which the limits of uncertainty of the measuring instrument are specified (the capability of the instrument).
Mechanical Completion	The stage when the construction and pre-commissioning of systems and plant are successfully completed in full compliance with the COMPANY technical specifications and in accordance with procedures and included checklists.
Non-Conformity Report	Document recording a non-fulfilment of a requirement, and the related corrective action(s) taken in place from CONTRACTOR in order to solve as soon as practicable the occurred non-conformity.
On-Line Instruments	Instruments connected to process and utility lines or equipment via primary isolation valves.
Process Owner	The CONTRACTOR representative in charge for the management of a process within the Project Management Team Organization and related P-QMS, and/or for a resolution of a dedicated action within the Project Organization
Product Category	The Product Category is focused on the equipment and material characteristics that may vary project by project due to the complexity of the project and the technology used.
Project	Front End Engineering Design for Gas Gathering Pipelines (Phase1)
Quality Assurance	Part of quality management focused on providing confidence that quality requirements will be fulfilled.

Quality Assurance Specification	CONTRACTOR document specifying the minimum requirements related to subcontractors QMS and QA/QC Manual specific for the works
Quality Control	Part of quality management focused on fulfilling quality requirements in terms of control and monitoring of the respect of technical requirements defined in CONTRACTOR scope of work.
Quality Control Specification	CONTRACTOR specification for Quality Control requirements. This standard compiles in Annex-1 all applicable ITS(s) and relevant Inspection and Test Report Forms
Quality Management System	Part of the organization's management system that focuses on achievement of results, in relation to the quality objectives, to satisfy the needs, expectations and requirements of contractor, company and of any other interested parties, as appropriate.
Shall	The word 'shall' indicate a mandatory requirement.
Should	The word 'should' indicate a recommendation.
SPDC	Shell Petroleum Development Company Nig. Limited
Subcontractor	SUBCONTRACTOR selected by CONTRACTOR to execute part of Engineering/Detail Design, Procurement and Construction
Technical Query	During detailed design and fabrication activities, technical questions and / or requests for clarification from specified requirements will be resolved between CONTRACTOR and COMPANY using the agreed query form. Subcontractors, Vendors and Suppliers shall systematically submit queries through CONTRACTOR for further treatment with COMPANY.
Third Party	An independent certification agency, accredited from a recognized International Forum of Associations, related to the object of the intervention, rather than certification.
Vendor	Organization, including its sub-vendors that sells, manufactures, fabricates or otherwise produces or supplies equipment and/or goods and services for project.
Work	All work and services to be performed and goods, services and other things to be provided by CONTRACTOR or SUBCONTRACTORS to design, procure, construct, erect and install the installations and test the systems and plant to satisfy all of contractor's obligations under the Agreement.

Abbreviations

Abbreviation	Meaning
BFPCL	Brass Fertilizer & Petrochemical Company Limited
CAR	Corrective Action Request
CPY	Company
CTR	Contractor
EAL	Epic Atlantic Limited (FEED Contractor)
EPCIC	Engineering, Procurement, Construction, Installation and Commissioning
FAT	Factory Acceptance Test
HP	Hold Point
ITP	Inspection and Test Plan
KPI	Key Performance Indicator
NACE	National Association of Corrosion Enterprises
NCR	Non-Conformity Report
PAR	Preventive Action Request
PIL	Project Inspection Leader
PMR	Project Management Review
PO	Purchase Order
PQP	Project Quality Plan
PQR	Procedure Qualification Record
PSL	Product Specialist Leader
PSR	Project Status Report
PVT	Performance Verification Test
QA	Quality Assurance
QC	Quality Control
QAS	Quality Assurance Specification
QCS	Quality Control Specification
QMS	Quality Management System
Shell	Shell Petroleum Development Company

SPC	Project Specification
SPDC	Shell Petroleum Development Company
TBE	Technical Bid Evaluation
TP	Third Party
VEMI	VENDOR Management
WBS	Work Breakdown Structure
WP	Witness Point
WPS	Welding Procedure Specification

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

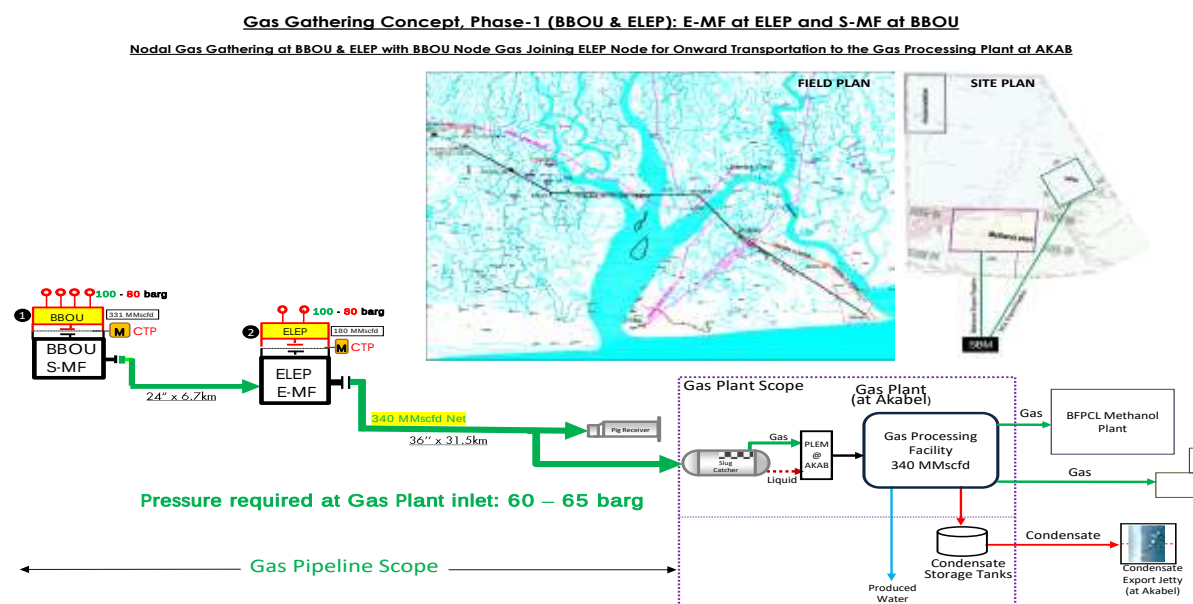


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This document streamlines the construction plans and practices for the pipeline construction /laying services for the evacuation and transportation of wet gas from SPDC wells at Bobouwe and Elepa fields to the Akabel gas treatment plant at the fertilizer processing plant for the phase 1 bulk lines and pipelines, applied for the realization of SPDC-BFPCL upstream gas supply project.

In particular, it provides fleet of construction activities plans involving civil works, excavation or HDD installation practices, fabrications and the steps in the installation of the pipelines, pre-commissioning & commissioning and covering plans on the following:

- Preliminary -start up Plan Process
- Planning for each of the Activities
- Planning Construction Methods
- Planning for construction, equipment and machinery
- Procurement of materials
- Planning for organization and employee skills
- Planning for required documents and drawings
- Financial planning

The plan shall allow for creation of the road map, and organizational structure (functions and responsibilities) to be implemented for the project, the design, planning, control and improvement of the operational processes to be applied to the activities in order to enable the organization to satisfy the COMPANY (or customer's) needs, as well as applicable statutory and regulatory requirements.

Overall, it will ensure critical actions are accurately planned and communicated to the teams, and stakeholders and ensuring that they delivered on target. It also provides useful insights into the timelines, progress and statuses of the multi projects processes of contracting and sub- contracting. Again, it reveals the potential roadblocks and the risks that may hinder the progress of the initiatives of the best practices to be used for effective pipeline project plan, planning tools to be used and ensuring that all key areas (auditing, technical aspects, performance tests all liabilities and level of liquidated damages) of the contract are addressed including quality plan.

1.3 Scope

The scope of the document shall incorporate the lists of goals, deliverables, features, Functions tasks, deadlines and cost. It constitutes the overall needs of the pipelines' as well as detailing the

benefits amongst others the milestones, that will be tracked from inception to the completion of the pipeline construction project, It will extend to document, WBS, communication plan, visual plans and rendering the construction specifications, drawings etc for sharing with stakeholders, government and local officials etc.

2. Regulations, Codes And Standards

Document Number	Title
ANSI/ASME B31.8	Gas transmission and distribution piping systems
API 104	Welding of pipelines and related facilities
Pipeline act Chapter 338	Laws of the federation of Nigeria 1080
DPR (NUPRC) OGISP	Guidelines and requirements for the application of oil and gas industry permits
ISO 9001 ed. 2015	Quality management systems – Requirements
ISO 10005:2005	Quality Management – Guidelines for Quality Plans

The latest versions of the regulations, codes, and standards and extant amendment or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Pipeline Construction Plan

3.1. General

Quality standards, performed within the network critical paths, without delay in timeline, within budget, to international standards. This is executed to the full satisfaction of the client and stakeholders.

It stems from the design output and information by EAL, which shall be sequenced using Gantt Chart and other approved tools by BFPCL.

It allows for the following:

- The Proper Monitoring of Progress and Reduce the Repeated Works.
- The Work Breakdown Structure (WBS)
- The Improvement of Open Communication, Queries / Answer Tracking and Conflict Resolution
- The Improvement of Tracking Individual Tasks of Team Members and their Levels of Accomplishment
- The Improvement of Reporting
- The Improvement of Risks Management Plan, Safety Management Plan and Security Plan,
- The Maximizing of the Efficiency of Resources by Getting the Best Labor, Materials and Equipment.

3.2. Preliminary Requirements to Develop The Construction Plan

The development of a proper pipeline construction plan may depend on the contract type -. : Turn-Key Contract, BOT, EPC and a single construction contract.

Irrespective, the responsible persons or contractor shall get themselves acquainted and familiarize with the followings to carry out the initial desk top studies and review of the relevant information and documentations:

- The ROW and Survey Works.
- The Physical Factors and Related Human Factors.
- The Environmental (EIA), and Political Factors.
- The Access to Transport Heavy Materials, Heavy Equipment and Personnel to Site of Construction.
- The Types of Construction Obstacles to be Encountered.
- The Topographical and Geotechnical Analysis - like for drilling for HDD.
- Jurisdictions- FG, SG, LG, Legal Authorities and Approval / Permits and Licence.

3.3 Organisation

The pipeline contraction project manager is responsible to assemble the right team with good experience and skillset. He is also responsible to deploy the professionals to their Work Breakdown Structure to deliver construction of the pipeline with required quality, on target with cost effectiveness.

The project manager shall work with the planning team whom shall be provided with the facilities and software tools to produce the Gantt Chart, timelines of different activities and the complete works from the inception to the completion of the contract.

The planning tools and software must be agreed and approved by BFPCL.

4. Pipeline Construction Plan Methodology

4.1. General Plan Methodology

Many international codes and standards are required for construction practices. To make a construction plan you need to identify all aspects of the project including the project management team, stakeholders, activities, the resources requirements, schedules and budget. The following are required for the implementation of the methodology

- The Definition of needs
- The Materials Take off (MTO) plan
- Identify Legal Frameworks and Pipeline Construction Permission
- A Full Set of Documentations and AFC drawings
- The Technical Aspects
- The Employee Obligations
- Rely on Information
- Performance Test
- All Limits on Liabilities
- Level of Liquidation Damage.

The methodology for making construction plan is similar to making a plan for any type of project. However, while the basic structure is the same, the nature of the construction requires specific Software that must be approved by BFPCL.

The methodology of pipeline construction is built from five phases.

Initiation: Determine the stakeholders, resources and project budget

Planning; Create Specific, measurable, attainable, realistic and timely goals. Define the work activities and create the schedule.

Execution: Assign Contractors team tasks and oversee project

Monitoring and Control: Monitor and track progress

Control of Close out Project and Evaluation Performance

The construction manager will now combine all the issues to produce the plan

4.2. Legal Framework and Compliance with Authorities

The contractor shall familiarize itself with the regulations, authorities and shall compile with any applicable requirement of Federal Government, State Government, Local Government and

individual and community requirements. Below, are some of the referenced regulations;

- **DPR (NUPRC) -OGISP** : Guidelines and Requirements for Application of Oil and Gas Industry.
- **Oil Pipeline Act Chapter 338** : Laws of the Federation of Nigeria
- **The Petroleum Act of 1969**
- **Mineral oils (Safety) regulation**
- **FEPA Act** – Federal Environmental Guidelines
- **EGASPIN** – ENVIRONMENTAL Guideline and standards for the Petroleum industries in Nigeria
- **STATUTORY** compensation in oil and gas industries
- **Oil Pipeline Act** mineral and mining with its incidental regulations
- **NSITF** Insurance Trust Fund
- **Bayelsa State** Oil & Environmental Commission
- **Bayelsa regulation**
- **OPL Act license 2004 Law of Nigeria**
- **Bayelsa State Environmental and Development Planning Authority Law 1998** under section 4(2), for a permit to survey (PTS) a pipeline route to be issued

4.3. MTO Plan

Once the contractor defines the material takeoff, the contractor can prepare a construction estimate. This is done by assigning cost to the itemization construction materials that were previously identified in the material takeoff (MTO)

Strategy will be to develop and follow, consistent projects contracting, commercial and other related strategies.

A sound administration and filing system will be set up to support material take-off and specifications, - tracking and managing process to ensure timely deliveries.

A Quality Management programme will be developed and followed for the manufacturing and fabrication to ensure compliance with codes and standards.

Expediting plan will focus on interface coordination, planning and scheduling, tracking, compliance monitoring and change management, control and verification, documentation and dispatch clearance.

Proper attention will be given to functional specification, quality, price, spares, FAT, vendor representation, and timely delivery, - as part of total cost of ownership concept.

4.4. Execution Strategy Plan for Pipeline Construction and Installation

Work Dossiers will be developed & maintained, covering the following:

- Project Plans
- Procedures & Methods
- ITPs & JHAs
- Work Packs
- Pre-Commissioning Loop Packs
- Pre -Start-up & Commissioning Checks
- Work Packs will be developed as instruments of work instructions for the various subsystems, covering the following key aspects:
 - ⇒ Activity Overview
 - ⇒ Scope of Work
 - ⇒ Method Statement
 - ⇒ Job Hazard Analysis
 - ⇒ Inspection & Test Plan (ITP) / Inspection & Test Records (ITR)
 - ⇒ Weld Fabrication Data (including NDT)
 - ⇒ Hydrotest Report
 - ⇒ Certificates (Discipline Acceptance & Work Completion Certificates)
 - ⇒ Technical Query / Deviation Request / NCR / CAR
 - ⇒ Punch List
 - ⇒ Material Receipt & Inspection Report
 - ⇒ Installation Drawing / As-Build Drawing
 - ⇒ Pre-Commissioning Overview
 - ⇒ Progress Tracking & Reporting
- Loop Pack will be developed for each sub-system, covering:
 - ⇒ System Limit PEFS Drawings
 - ⇒ ILD – to guide both the cold loop checks, and functional tests.
 - ⇒ Check Sheets & Test Forms
- The Procedures, Method Statements, JHA, & ITPS will cover scopes and activities, - examples of some key relevant scopes/activities below:
 - ⇒ Lifting & Erection / Installation Works
 - ⇒ Lifting Equipment Inspection, Testing, & Colour Coding
 - ⇒ Mobile Plant
 - ⇒ Confined Space
 - ⇒ Scaffolding, Ladder, & Rope Access Safety
 - ⇒ Welding (including Specifications, Qualifications, Consumables, NDT)
 - ⇒ Piping Fabrication & Installation (including Testing)

- ⇒ Blasting & Painting
- ⇒ Electrical Works
- ⇒ Instrumentation Works
- ⇒ Materials Handling & Storage
- ⇒ Safe Handling of Chemicals

4.5. Planning of Changes

Changes to Project Quality Management Systems shall be managed considering their potential consequences, the maintenance of system integrity and the allocation or reallocation of responsibilities and authorities to available resources.

Such changes can result from:

- Implementation of corrective and/or preventive actions.
- Management review of Project Quality Management System (PQMS).
- Implementation of lessons learned.

5. Inspection Test Plan, WPS and PQR

The Table is the welding inspection test plan.

Table 1 the description, welding, project procedure,

OP N°	DESCRIPTION OF INSPECTION ACTIVITY	PROJECT WELDING PROCEDURE	INSPECTION FREQUENCY	ACCEPTANCE CRITERIA
1	SAFETY (JSA)	Safety Manual	Daily	Ensure ALL employees are wearing proper PPE for the relevant tasks at hand
2	VERIFY WELDING PROCEDURE	Project Welding Manual	As Needed	Applicable Weld Procedure Correct WPS and PQR for weld
3	VERIFY QUALIFIED WELDERS	Operator Specification	- Prior to Welding - Change of Welder and/or Procedure	- Welder Qualification Log - Verify Operator Qualification requirements (as applicable)
4	PIPE INSPECTION	Internal Specification	Each Pipe Identification and Joint condition	Pipe clearly identified and free from rusting and debris
5	BEVEL PREPARATION	Weld Procedure	Each Joint	Clean bevel as per Welding procedures
6	FIT-UP PROCESS	Weld Procedures	Each Joint	- Root Gap - Root Face - High-low - Weld Procedures
7	PRE HEATING	Weld Procedure	Each Joint	Minimum as specified and verified by temp stick or equivalent
8	INTER-PASS TEMPERATURE	Weld Procedures	Each Joint	Minimum as specified and verified by temp stick or equivalent
9	TRAVEL SPEED	Weld Procedures	As Needed	Range is inches/min per WPS/PQR
10	WELDING PARAMETERS (AMP, VOLT, e.t.c)	Weld Procedures	As Needed	Range is amps per WPS/PQR
11	VISUAL INSPECTION	Weld Procedures	Each joint	Welding Standard Ref
12	NON-DESTRUCTIVE TESTING	NDE Procedures	As agreed	ASME V

13	REPAIR OF DEFECTS (CRACKS, ARC STRIKES)	Repair Procedure	As Needed	Welding approval Inspector
14	FIELD TRANSITIONS	Internal Speculation	As Needed	Customer Specification
15	WELD ACCEPTANCE	Weld Procedures & NDE Procedure	Each Weld	• API 1104 • B31.8 • NDE Procedures
16	CHECK CONSUMABLES AND WELDING EQUIPMENT	Operator Specification	Daily	• Proper consumable storage • Welding equipment in proper condition

Welding Procedures are the guidelines used to perform a weld. They are designed to provide a record of the welding variables used and the inspection results obtained during the procedure qualification test. They can also provide the instructions for the welder to use in production in order to produce acceptable welds. Usually welding procedures are developed in accordance with a welding code or standard, and with few exceptions*, require that physical weld samples be produced, inspected, and tested to establish qualification. Welding procedures are usually divided into two categories, the Procedure Qualification Record (PQR) and the Welding Procedure Specification (WPS).

Procedure Qualification Records are the documented values used during the actual welding test and all the inspection and test results obtained from the actual test samples.

Welding Procedure Specifications are usually documented work instructions that can be used by the welder to conduct welding operations, and are based on, but not necessarily the same as, the parameters used for the Procedure Qualification Record.

We will consider the Procedure Qualification Record and the inspection and testing performed during its qualification.